



The Last-Mile Audit

Are we failing specific regions, or is this a nationwide problem?

Delivery performance vs. customer sentiment · Olist marketplace orders, 2016–2018

A regional problem, driven by mis-calibrated promises



6.8%

of delivered orders
arrive after the
promised date



12.7%

late rate in the North-
East, about double the
South-East and South



1.74★

average review for
orders more than 5
days late, vs. 4.29★
on time



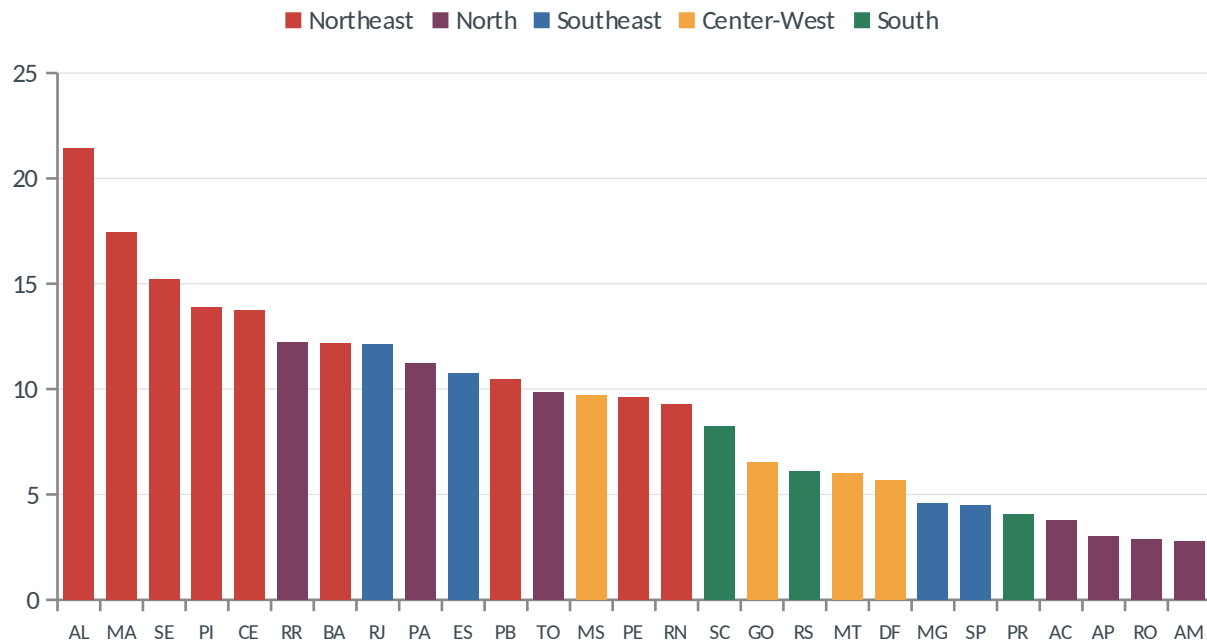
72%

of late orders were
handed to the carrier
on time; delay builds
in transit

Answer to the CEO: it is regional. The North-East is late 12.7% of the time vs. about 6% in the South-East and South, because its estimates carry too little buffer.

Late deliveries concentrate in the North-East and Rio

% of delivered orders that arrived late, by customer state



● **North-East: 12.7% late**

vs. 6.1% in the South-East and 5.9% in the South.

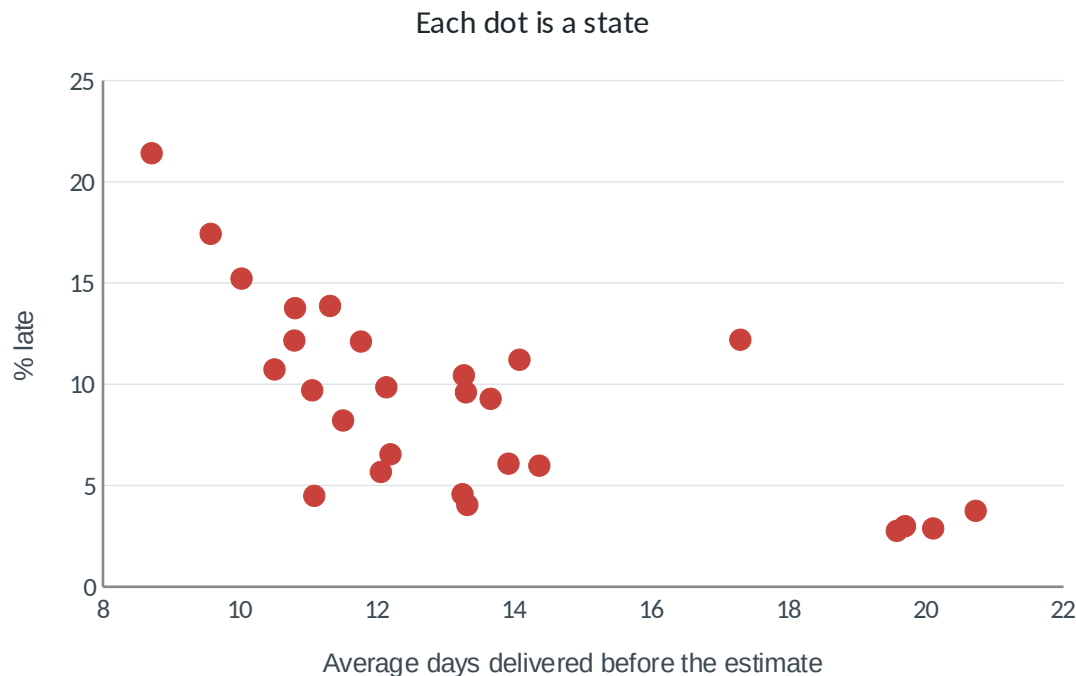
● **Worst: AL 21%, MA 17%**

Alagoas and Maranhão: 4–5× the São Paulo rate of 4.5%.

● **Rio de Janeiro: 12.1%**

Far above other South-East states, even though most sellers are nearby.

Thin estimate buffers, not distance, drive lateness



Far North: ~20 days early

Amazonas, Amapá, Rondônia, Acre are the most remote, yet only about 3% late.

North-East: ~9–11 days early

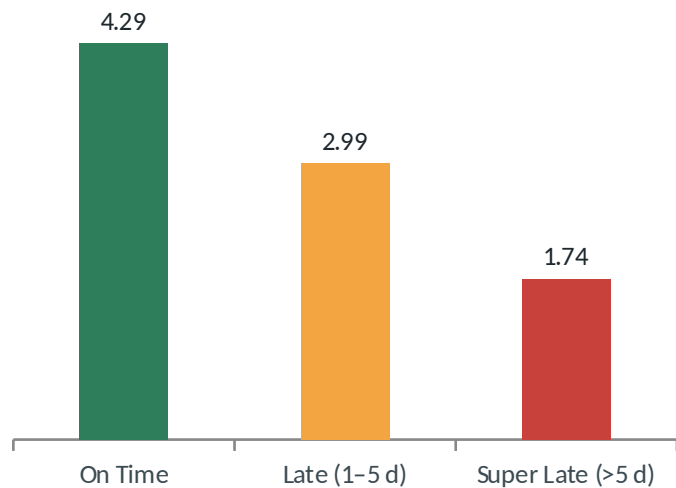
Alagoas, Maranhão, Sergipe get the least buffer and are 15–21% late.

Distance $\rho = +0.06$; buffer $\rho = -0.70$

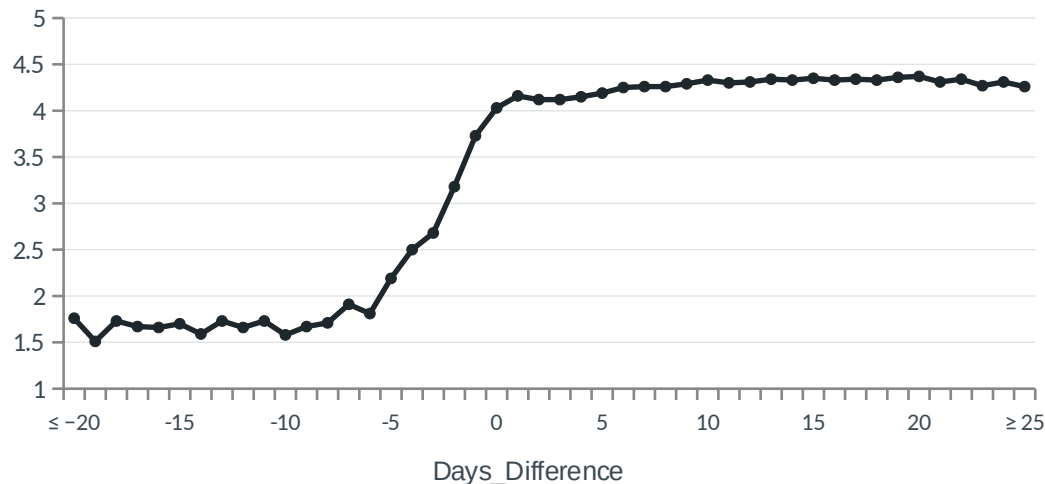
Across 27 states, the buffer explains lateness and distance doesn't.

A broken promise costs two stars

Average review score (1-5)

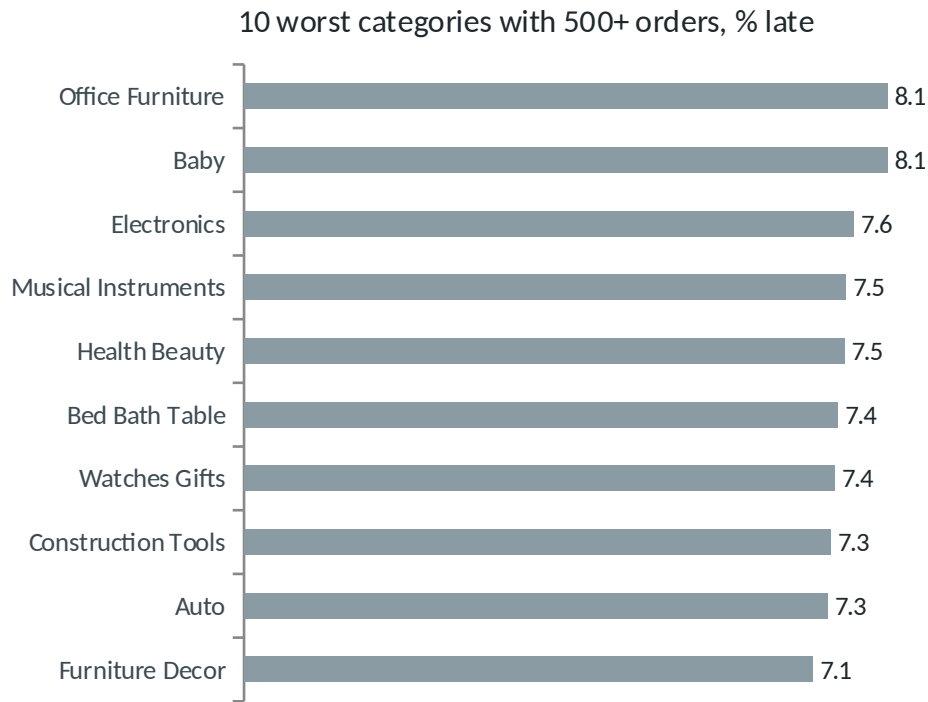


Avg review by Days_Difference (negative = days late)



Arriving early doesn't earn extra stars, but the score falls sharply from the first day late and keeps falling for about a week.

Destination matters far more than product category



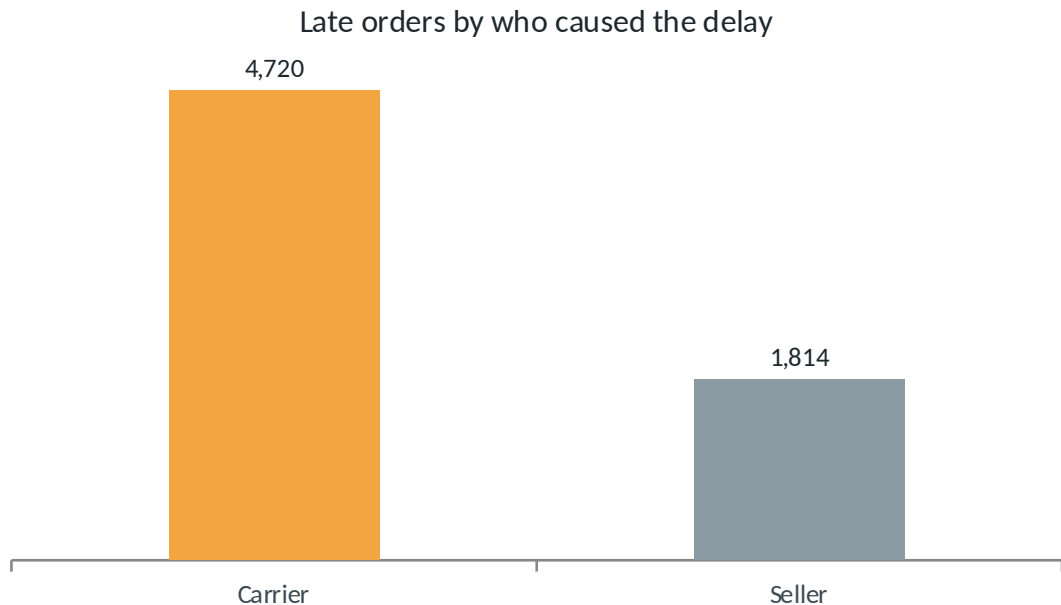
Spread of late rates

7-8% 10 worst categories

3-21% across states

Even the worst categories are only a point or so above the 6.8% average. Furniture is about as punctual as electronics (Furniture Decor 7.1% vs. Electronics 7.6%).

Who to fix: 72% of delays build after seller hand-off



72%

of late orders left the seller on time:
carrier / last mile

28%

started with the seller missing its
shipping deadline

Each late order is assigned an owner: seller if it reached the carrier after `shipping_limit_date`, otherwise carrier / last mile.

Four actions to fix the last mile

1

Recalibrate estimates by region

Add buffer to North-East and Rio routes (today about 9–12 days early). Trim the ~20-day padding in the far North.

2

Carrier scorecards in NE and Rio

72% of delays happen in transit. Renegotiate or re-route the worst lanes.

3

Seller hand-off deadlines

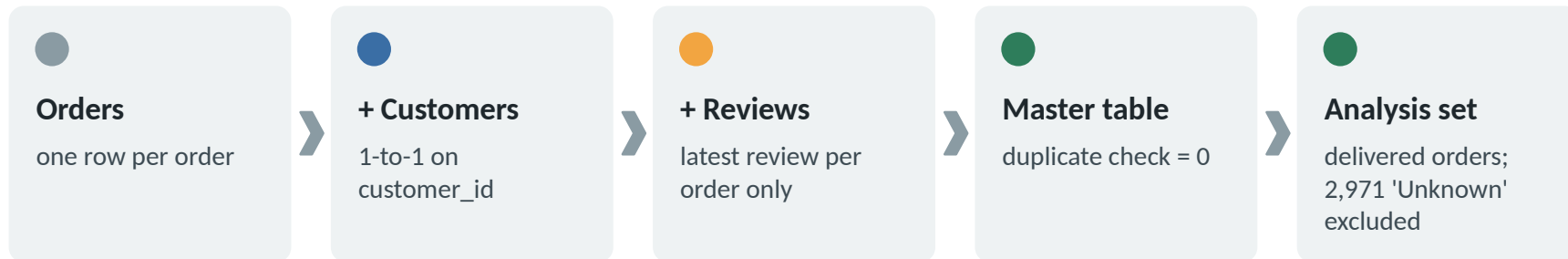
28% of late orders start with a seller missing its shipping deadline. Flag repeat offenders.

4

Track late rate on the dashboard

Reviews drop from the first day late. Watch the worst states and message customers when a parcel runs late.

Appendix: how the data was built and cleaned



- Reviews de-duplicated before joining; the notebook prints 0 duplicate orders after the joins.
- Days_Difference = estimated – delivered date (day level). On Time ≥ 0 ; Late 1–5 days; Super Late > 5 days.
- 2,971 orders without a delivery (canceled, unavailable, still shipping) are labelled 'Unknown' and excluded.
- Distance: haversine between the seller's and customer's zip-code centroids, averaged per order.